

Programming Languages -- Guided Notes ANSWER KEY

Unit tp-3.7 | CompTIA Network+ N10-009 Obj. FC0-U71 3.7 | N10-008 FC0-U71 3.7

For instructor use / distribute after students complete the notes.

Question 1

A programming language is a formal language used to write _____ that computers can execute. Source code is written in _____ -readable text, then converted to machine code. The two main conversion methods are _____ (converts all code at once before running) and _____ (converts and runs line by line).

Answer: Instructions (programs); human; compilation; interpretation.

Question 2

_____ code (binary machine instructions) is what the CPU executes directly. _____ language uses mnemonics (like MOV, ADD) to represent machine instructions. High-level languages like Python and Java are _____ readable and must be compiled or interpreted before the CPU can run them.

Answer: Machine; assembly; human.

Question 3

Python is an _____ language known for readable syntax and use in data science, automation, and web development. Java is a _____ language that compiles to bytecode run on the _____ (JVM), enabling write-once run-anywhere portability. JavaScript runs primarily in _____ to make web pages interactive.

Answer: Interpreted; compiled; Java Virtual Machine; browsers.

Question 4

C and C++ are _____ -level languages often used for _____ programming, game engines, and performance-critical applications. They give programmers direct access to _____ management, which increases performance but also increases the risk of _____ vulnerabilities like buffer overflows.

Answer: Low (systems); systems-level; memory; security.

Question 5

SQL (Structured Query Language) is used to _____ and _____ data in relational databases. HTML (Hypertext Markup Language) defines the _____ of web pages. CSS (Cascading Style Sheets) defines the _____ (colors, fonts, layout). Neither HTML nor CSS is technically a _____ language -- they are markup/styling languages.

Answer: Query; manage (create, update, delete); structure (content); appearance; programming.

Question 6

A scripting language (Python, Bash, PowerShell) is typically _____ and used for _____ tasks. A PowerShell script can automate _____ administrative tasks like creating user accounts. A Bash

script on Linux can automate _____ and file operations. Scripts reduce _____ compared to doing tasks manually.

Answer: Interpreted; automation; Windows; backups; time/effort.

Question 7

An IDE (Integrated Development Environment) combines a _____ editor, _____ (finds syntax errors), _____ (automates build/run), and _____ (steps through code to find logical errors) into one tool. Popular IDEs include Visual Studio Code, PyCharm, and _____.

Answer: Code; linter; build tool; debugger; IntelliJ (or Eclipse).

Question 8 -- Real World Application

REAL WORLD: A student wants to build a personal website, create some scripts to automate their homework file organization, and eventually get into data analysis. They ask you which programming language they should learn first. Recommend one language, give three reasons it is the best starting point for their goals, and name one limitation.

Answer: Recommendation: Python. Reason 1 -- Beginner-friendly syntax: Python reads like plain English, uses indentation instead of excessive punctuation, and has gentle error messages. It is the most commonly taught first language in universities and bootcamps. Reason 2 -- Covers three of their goals directly: Python can build web apps (Flask, Django), write automation scripts (file management, renaming, sorting), and is the dominant language in data analysis (pandas, NumPy, matplotlib). Reason 3 -- Huge community and library ecosystem: beginner stuck on any problem can find dozens of tutorials. Limitation: Python is interpreted and not the best choice for performance-critical applications or front-end web interactivity -- for the actual website UI they will also need to learn some HTML/CSS and possibly JavaScript.